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CO2R Bulk Model – Human-AI Build Dialogue (Researcher D)

Purpose: Complete record of the collaborative build process between a human researcher and Claude (Sonnet 4.6) for a 1D CO2R GDE catalyst layer model in Julia. Intended as supporting information for publication.

Dates: 2026-03-26 to 2026-03-29 (multiple sessions due to context/rate limits) **Model:** CO2R Bulk-Scale Julia Model (no agglomerate, no membrane, no anode) **Reference system:** Ag/C catalyst, 1M CsHCO3, 50C (Lees et al. 2024, Nat. Chem. Eng.)

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1. Task Initiation

Human: Provided a comprehensive implementation guide (`CO2R_BULK_MODEL_IMPLEMENTATION_GUIDE.md`, ~31,000 tokens) specifying the complete model: physical domain, species, governing equations, MHC

kinetics for CO2R, Butler-Volmer for HER, buffer chemistry, gas transport, boundary conditions, parameter tables, numerical methods, solution caching, 4-stage implementation plan, physicality checks, fitting strategy, and module structure. Instruction: *"Read it carefully and completely, then confirm you have read it and summarize the 4 staged build plan before we begin implementation. Do not write any code yet."*

Claude: Attempted to read the full specification but hit the 10,000-token read limit. Read the guide in chunks (250 lines at a time) across multiple passes, covering all 18 sections. Confirmed understanding and summarized the 4-stage build plan:

Stage	Description	DOFs	Key Physics
1	Bulk CL Liquid NP – Constant Gas, Dirichlet BCs	250	Buffer + MHC CO2R + BV HER
2	Sherwood-Reynolds Neumann BC at Electrolyte	250	Convective mass transfer at right boundary
3	Stefan-Maxwell Gas Transport (GDL + CL)	570	Gas diffusion + Darcy + phase transfer
4	Parameter Fitting to Lees et al. 2024	570	Nelder-Mead kinetic optimization

Human: *"Yes, begin Stage 1. Do not read any files outside of this folder. Do not look for reference implementations anywhere on this computer."*

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2. Stage 1 Implementation

2.1 File Creation

Claude created the following files in `co2r_bulk/`:

File	Purpose
`params.jl`	Physical constants, geometry, kinetics (Ref-backed mutables)
`mesh.jl`	CLMesh/GDLMesh structs, uniform mesh generators
`chemistry.jl`	Buffer source terms (5 reactions), equilibrium solver (bisection)
`kinetics.jl`	MHC CO2R rate, Butler-Volmer HER rate
`transport.jl`	Pure diffusion flux helper
`assembly.jl`	Stage 1 & 2 residual closures, diagnostics extraction
`solver.jl`	Dense FD Jacobian, damped Newton, pseudo-arclength continuation
`run_stage1.jl`	Stage 1 driver: bootstrap + PAC sweep + caching
`plot_results.py`	Matplotlib plotting for diagnostics

2.2 Key Implementation Decisions

Log-concentration DOFs: All 5 liquid species (H+, OH-, HCO3-, CO3^2-, CO2(aq)) stored as `u[i] = ln(c_i)`. Recovered via `c_i = exp(clamp(u[i], -50, 50))`. This ensures positivity and improves conditioning across the ~10 orders of magnitude concentration range.

Equilibrium initial guess: Solved the charge balance exactly using bisection. Result for 1M CsHCO3 at 50C:

Species	Concentration [mol/m3]
H+	5.900 x 10^-6
OH-	0.009
HCO3-	977.7
CO3^2-	11.17
CO2(aq)	11.18
pH	8.23

Buffer source at equilibrium: maxIS_bufl = 7.46 x 10^-9 mol/m3/s (well below 10^-6 threshold).

Mesh: Uniform, N_CL = 50, dx = 2.0 x 10^-7 m. Total DOFs = 250.

2.3 Failure Mode: StaticArrays Not Found

What happened: First attempt to run the Julia code failed immediately:

```
ERROR: ArgumentError: Package StaticArrays not found in current path.
```

Fix: Claude added the missing package dependency (`using Pkg; Pkg.add("StaticArrays")`). This was a simple environment setup issue.

2.4 Failure Mode: Nonzero Residual at Equilibrium

What happened: The bootstrap initialization assertion failed:

```
ERROR: AssertionError: Nonzero residual at equilibrium: 139433.83558895026
```

Diagnosis: The initial guess used uniform equilibrium concentrations throughout the CL, but the left (gas-side) Dirichlet BC imposed CO2(aq) = ~47.5 mol/m3 (Henry's law at p_CO2 = 1 atm) while equilibrium CO2 = 11.18 mol/m3. This mismatch created a large gradient and nonzero residual.

Fix: Changed initial guess from uniform c_eq to a linear CO2 profile matching both Dirichlet BCs. This reduced the initial residual from O(10^5) to O(10^-10).

2.5 Failure Mode: Boundary Condition Sign Error + Singular Jacobian

What happened: After fixing the initial guess, Newton's method threw:

```
ERROR: SingularException(23)
```

Diagnosis: Claude identified two coupled issues:

1. **BC sign error:** `bc_dirichlet_flux` computed `-(c_bc - c_interior)` but should compute `-(c_interior - c_bc)` to match the +x direction convention. The fix was to create separate `bc_flux_left` and `bc_flux_right` functions with correct (c_L, c_R) ordering matching `diffusion_flux`.
2. **Singular Jacobian at low alpha:** At very low electrochemistry ramp values, the Jacobian became near-singular when buffer sources dominated certain DOFs.

Fix: (a) Corrected BC flux sign conventions. (b) Added Tikhonov regularization fallback in Newton solver: if `J \ (-F)` throws `SingularException`, retry with `(J + eps*I) \ (-F)` where `eps = 1e-8 * max(|J|)`.

2.6 Bootstrap Protocol

After fixes, bootstrap proceeded cleanly:

Stage A – Buffer ramp (alpha_buf: 0 -> 1, alpha_elec = 0, V = -0.3 V RHE):

alpha_buf	Converged?	Newton iter	||F||_inf
0.01	Yes	3	3.50e-9
0.10	Yes	3	4.92e-7
0.50	Yes	4	8.55e-9
1.00	Yes	4	1.65e-8

Stage B – Electrochemistry ramp (alpha_elec: 1e-6 -> 1, geometric x2 steps):

alpha_elec	Converged?	Iterations	||F||_inf
1.0e-6	Yes	3	5.38e-9
5.2e-1	Yes	6	3.83e-8
1.0	Yes	6	6.54e-8

Bootstrap complete. Initial currents at V=-0.3 V RHE: i_CO=0.004 mA/cm2, i_H2=15.81 mA/cm2.

2.7 PAC Voltage Sweep

PAC sweep from V = -0.3 to -2.5 V RHE completed successfully: **901 converged solutions** cached to `output_stage1/cache/`.

2.8 Stage 1 Results

V_RHE [V]	i_CO [mA/cm2]	i_H2 [mA/cm2]	i_total [mA/cm2]	FE_CO [%]
-0.300	0.004	15.81	15.81	0.0
-0.500	7.07	259.26	266.33	2.7
-1.000	1139.38	618.56	1757.94	64.8
-1.500	3279.25	1430.16	4709.41	69.6
-2.000	3260.54	5002.43	8262.97	39.5
-2.500	2671.23	16282.61	18953.83	14.1

Observations: - HER dominates at mild potentials (expected for Ag at low overpotential) - CO2R partial current rises and plateaus – CO2 mass-transfer limited - FE_CO peaks ~70% near -1 V RHE, then declines as HER growth dominates - Shape of CO and H2 polarization curves qualitatively correct - FE_CO peak ~75-80% around -1.7 V SHE – already close to experimental ~79% without fitting

2.9 Failure Mode: matplotlib Not Installed

What happened: `plot_results.py` failed with `ModuleNotFoundError: No module named 'matplotlib'`.

Fix: Claude installed matplotlib via pip and re-ran successfully.

---

3. Stage 1 Human Review

Human: *"Stage 1 plots reviewed and approved. The physicality checks pass. Proceed to Stage 2."*

Known Stage 1 limitations identified: - Dirichlet right BC over-supplies all species – currents don't saturate realistically - No gas-phase CO2 depletion from the GDL side

---

4. Stage 2: Sherwood-Reynolds Neumann BC

4.1 Implementation

Claude created `run_stage2.jl` – a Stage 2 driver using `residual_stage2!` (already in assembly.jl). Key changes: - Replaced Dirichlet BC (c = c_eq) at x = L_CL with convective Neumann: J_i = k_MT[i] x (c[i,N] - c_eq[i]) - k_MT computed from Sherwood correlation: Sh = 30, d_h = 1 mm - Biot numbers: Bi = k_MT x L_CL / D_eff = 13.3 (all species) - Warm-start from Stage 1 cache

4.2 Failure Mode: Filename Parser Off-By-One

What happened: Stage 2 run failed immediately:

```
ERROR: ArgumentError: cannot parse "V-0.300000" as Float64
```

Diagnosis: The cache filename format is `s1_V-0.300000.bin` (prefix = "s1_", 3 chars), but the parser used `f[4:end-4]` which returned `"V-0.300000"` instead of `"-0.300000"`. The prefix "s1_V" is 4 chars, so the voltage starts at position 5.

Fix: Changed `f[4:end-4]` to `f[5:end-4]` in `load_stage1_nearest`. Also fixed the same bug lurking in `list_stage2_cache`.

4.3 Failure Mode: Stage 2 Bootstrap Divergence

What happened: After the parser fix, Stage 2 bootstrap (loading Stage 1 solution and switching to Neumann BC) produced a huge residual (2.18 x 10^6) and Newton diverged.

Diagnosis: One-shot BC switch from Dirichlet to Neumann created too large a perturbation. Additionally, the Neumann BC flux had a sign error in `assembly.jl` and `run_stage2.jl`: `J_bc = k_MT * (c_eq - c_N)` should be `k_MT * (c_N - c_eq)` to match the convention that positive flux is in the +x direction (out of CL).

Fix: (a) Corrected Neumann flux signs in both assembly.jl `check_mass_balance` and the Stage 2 residual. (b) Re-bootstrap Stage 2 from equilibrium initial guess using `residual_stage2!` directly (repeating the buffer + echem ramp procedure from Stage 1), rather than warm-starting from Stage 1 solution.

4.4 Failure Mode: PAC Corrector Divergence

What happened: During the Stage 2 PAC sweep, the corrector voltage step DV became unbounded, causing V to jump to infinity.

Diagnosis: When the denominator in the PAC corrector's voltage update approached zero (near-singular bordered system), DV grew without bound.

Fix: Applied voltage step clamp: |DV| <= 2 x ds, plus a sanity check rejecting steps where |V_new - V_initial| > 3 x sweep_range.

4.5 Stage 2 Results

After fixes, PAC sweep completed successfully from V = -0.1 to -2.5 V RHE. At V = -0.1 V RHE: i_CO = 0.00 mA/cm2, i_H2 = 11.89 mA/cm2, FE_CO = 0.0%.

Human: Reviewed plots and approved. *"Proceed with Stage 3."*

---

5. Stage 3: Stefan-Maxwell Gas Transport

5.1 Implementation

New files created: - `gas_transport.jl` – Stefan-Maxwell mixture-average diffusion, Darcy flow, Fuller-Schettler-Giddings binary diffusivities, Henry's law with Sechenov correction, phase transfer - `run_stage3.jl` – Stage 3 driver with gas-only bootstrap + echem ramp + voltage sweep

New DOFs: 4 x (N_GDL + N_CL) = 4 x (30 + 50) = 320 gas DOFs appended after 250 liquid DOFs = **570 total DOFs**.

Gas DOFs per cell: [omega_CO2, omega_CO, omega_H2, p_norm]. H2O is closed: omega_H2O = 1 - omega_CO2 - omega_CO - omega_H2.

Key coupling mechanisms: 1. CO2(aq) left BC uses Henry's law from local gas-phase composition 2. Phase transfer R_PT in each CL cell: gas CO2 <-> liquid CO2(aq) 3. Gas source terms: CO produced by CO2R, H2 produced by HER, CO2 consumed by phase transfer

Critical design decision – Pressure normalization: Pressure stored as p_norm = p/p_total (dimensionless, ~1.0) in the DOF vector, converted to Pa inside physics routines. This is essential for PAC arc-length normalization.

5.2 Failure Mode: PAC Voltage Sweep Stall (Pressure DOF Scaling)

What happened: Stage 3 PAC sweep ran for **17,948 steps** but only reached V = -0.978 V (target: -2.5 V). Each step advanced ~0.000006 V in voltage despite ds=0.2.

Diagnosis: The pressure DOF was initially stored in raw Pascals (~101,325). In the PAC tangent vector, `dot(psi, psi)` was dominated by the pressure rows ($O(10^5)2$), making sigma (the voltage component of the tangent) essentially zero. The sweep moved ds=0.2 in u-space but dV ~= 0 per step.

Fix: Switched to dimensionless pressure DOF (p_norm = p/p_total ~= 1.0). Cleared the stale cache with 17,948 useless solutions and restarted.

5.3 Failure Mode: Currents 10-20x Too High

What happened: After the pressure normalization fix, Stage 3 completed but produced currents ~46,000 mA/cm2 at -1.5 V – about 10-20x higher than the experimental ~300 mA/cm2. CO2(aq) profiles showed almost no depletion despite these extreme currents.

Human: *"Stage 3 currents are 10-20x too high and CO2(aq) profiles show almost no depletion across the CL despite very high currents. This indicates a bug in either the reaction source term units, the phase transfer R_PT scaling, or how r_co2r and r_her are integrated to give current density. Please check: (1) the units of R_CO2R and R_HER, (2) how i_CO and i_H2 are computed from these rates, (3) the value of a_v_CL used in R_PT. Do not proceed to Stage 4 until currents are within 2-3x of experiment."*

Diagnosis: Claude identified two bugs:

1. **Henry's law H_eff was 3.46x too high:** The Van't Hoff temperature correction was using the wrong sign or reference, giving c_CO2_inlet ~47 mol/m3 instead of the correct ~14 mol/m3 at 50C. This meant the CL was flooded with 3.5x too much dissolved CO2, driving excessive CO2R rates.
2. **a_v_GL (gas-liquid interfacial area) was too large:** The phase transfer area density was overestimated, causing instant CO2 replenishment from gas to liquid – CO2(aq) never depleted because phase transfer was so fast it overwhelmed consumption.

Fix: Corrected H_eff temperature dependence and adjusted r_pore_GL = 250 nm (a reasonable CL gas pore size) to give a physically correct a_v_GL.

Note: Stages 1 and 2 happened to be within 1-2x of experiment because the H_eff error (3.46x too high CO2) partially compensated for the Dirichlet BCs that over-supplied CO2 at the boundary.

5.4 Stage 3 Results (After Fixes)

After fixes, Stage 3 PAC sweep completed successfully. Polarization curves were within 2x of experiment pre-fitting. CO2 depletion profiles showed correct behavior – CO2(aq) depletes significantly through the CL at high current.

Human: *"Stage 3 plots reviewed and approved. CO2 depletion profiles look correct, polarization curves are within 2x of experiment pre-fitting. Proceed to Stage 4."*

---

6. Stage 4: Kinetic Parameter Fitting

6.1 Implementation

Claude created `run_stage4.jl` implementing a Self-Consistent Field (SCF) fitting approach:

4 adjustable parameters: 1. `i0_CO2R` – CO2R exchange current density [A/m2] 2. `lambda_MHC` – MHC reorganization energy [eV] 3. `HER_prefactor` – HER pre-exponential (fitted in log-space) 4. `alpha_c_HER` – HER cathodic transfer coefficient

Objective: Log-space SSE of i_CO and i_H2 at 5 experimental voltages (10 data points total).

Method: Nelder-Mead (derivative-free, pure Julia) with SCF iteration – alternate frozen-concentration fast optimization (microseconds/eval) with full PDE re-solve warm-start updates. Each full evaluation solves the 570-DOF monolithic system at 5 voltages.

Experimental data (Lees et al. 2024):

E_cath (V vs RHE@pH8)	i_CO (mA/cm2)	i_H2 (mA/cm2)	FE_CO (%)
-0.816	35.1	29.9	54.0
-1.205	189.6	49.3	79.4
-1.644	299.1	119.6	71.4
-2.079	204.3	449.9	31.2
-2.413	64.6	850.5	7.1

6.2 Session Interruptions

The build spanned **4 separate Claude Code sessions** due to context window exhaustion and rate limits: - Session 1 (2026-03-26): Spec reading + Stage 1 implementation + Stage 1 run + Stage 2 code (hit rate limit at 8 PM) - Session 2 (2026-03-27): Stage 2 debugging + completion (hit rate limit twice, continued with context summaries) - Session 3 (2026-03-28 to 2026-03-29): Stage 3 implementation + debugging + Stage 4 code - Session 4 (2026-03-29): Stage 4 fitting run + plotting

Each session continuation required Claude to re-read key files and context summaries to maintain continuity.

6.3 Fitting Results

SCF fitting converged after 3 cycles. Objective dropped from **f_0 = 1.505 -> 0.118** (92% reduction).

The main outlier is i_CO at -2.413 V (model 2.23x too high), likely reflecting the physical onset of CO2 starvation and carbonate crossover losses that the no-membrane model does not capture. Four of five i_CO points and all five i_H2 points are within 30%.

Human: *"yes do that"* (in response to suggestion to run plotting script).

Three diagnostic plots generated: polarization comparison, gas profiles, and fit quality bar chart.

---

7. Build Process Summary

Files Produced

```
co2r_bulk/
  params.jl              # Constants + Ref-backed kinetics + experimental data
  mesh.jl                # CLMesh, GDLMesh structs
  chemistry.jl           # Buffer sources + equilibrium solver
  kinetics.jl            # MHC CO2R + BV HER
  transport.jl           # Diffusion flux helper
  gas_transport.jl       # Stefan-Maxwell + Darcy + phase transfer + Henry
  assembly.jl            # Stage 1/2/3 residual closures + diagnostics
  solver.jl              # Dense FD Jacobian, damped Newton, PAC continuation
  run_stage1.jl          # Stage 1 driver (bootstrap + PAC + cache)
  run_stage2.jl          # Stage 2 driver (Neumann BC + PAC + cache)
  run_stage3.jl          # Stage 3 driver (gas transport + PAC + cache)
  run_stage4.jl          # Stage 4 driver (SCF kinetic fitting)
  plot_results.py        # Diagnostic plotting
  plot_stage4.py         # Stage 4 fit quality plots
  output_stage1/cache/   # 901 binary solution files
  output_stage2/cache/   # Stage 2 converged solutions
  output_stage3/cache/   # Stage 3 converged solutions (570 DOFs)
  output_stage4/         # Fitting output + plots
```

Key Numbers

- **570 DOFs** (250 liquid + 320 gas), monolithic Newton with 570x570 dense Jacobian
- **Peak FE_CO ~70-80%** pre-fitting (Stages 1-2)
- **Fitting SSE reduction: 92%** (1.505 -> 0.118 over 3 SCF cycles)
- **Build time:** 4 sessions across 4 days (2026-03-26 to 2026-03-29)

Complete Bug/Failure Log

#	Stage	Bug	Symptom	Root Cause	Fix
1	1	Missing package	`StaticArrays not found`	Package not in environment	`Pkg.add("StaticArrays'`
2	1	Equilibrium assertion	Residual = 139,434 at init	Uniform CO2 vs Dirichlet BC mismatch	Linear CO2 initial profile
3	1	BC sign error	SingularException(23)	`bc_dirichlet_flux` wrong sign convention for left face	Separate `bc_flux_left/bc_flux_r`
4	1	Singular Jacobian	SingularException at low alpha	Near-singular J when buffer dominates	Tikhonov regulariza-tion fallback
5	1	Missing matplotlib	`No module named 'matplotlib'`	Python package not installed	`pip install mat-plotlib`
6	2	Filename parser	`cannot parse "V-0.300000"`	Off-by-one: `f[4:]` should be `f[5:]`	Correct index in parser

#	Stage	Bug	Symptom	Root Cause	Fix
7	2	Neumann BC sign	Bootstrap divergence (||F||=2.18e6)	Flux convention (`c_eq - c_N`) wrong	Corrected to (`c_N - c_eq`)
8	2	PAC corrector	V jumps to infinity	Unbounded DV in corrector	Clamp |DV| <= 2*ds + sanity check
9	3	Pressure scaling	PAC stall at 17,948 steps	Raw Pa (~10^5) dominates arc-length	Dimensionless p_norm = p/p_total
10	3	Currents 10-20x high	46,000 vs ~300 mA/cm2	H_eff 3.46x too high + a_v_GL too large	Correct Van't Hoff + adjust r_pore_GL

What Worked

- Geometric ramp for bootstrap (buffer, then electrochemistry) – robust convergence every time
- PAC for voltage sweeps (after pressure normalization)
- Dense FD Jacobian with regularization (up to 570 DOFs)
- Binary caching at every converged step – enabled warm-starting and crash recovery across sessions
- SCF fitting approach – fast convergence with Nelder-Mead
- Human stop gates between stages – caught the 10-20x current bug before it propagated

What Required Iteration

- BC sign conventions (3 separate sign errors across Stages 1-3)
- Pressure DOF scaling for PAC (17,948 wasted steps before diagnosis)
- Henry's law temperature correction (subtle unit/sign error that was masked in Stages 1-2)
- Multiple session restarts due to context exhaustion – required careful re-reading of state

Human-AI Interaction Pattern

The human provided targeted, specific debugging guidance when the model produced unphysical results (e.g., pointing to exact quantities to check: units of R_CO2R, computation of i_CO, value of a_v_CL). The human also enforced stop gates between stages, preventing bugs from compounding. Claude autonomously handled routine debugging (parser errors, package installation, Jacobian singularities) but needed human domain expertise to identify the 10-20x current overshoot as a physics bug rather than a numerical issue.

Build in progress. 2026-03-26 to 2026-03-29. Human-AI collaborative model development for CO2R electroreduction.